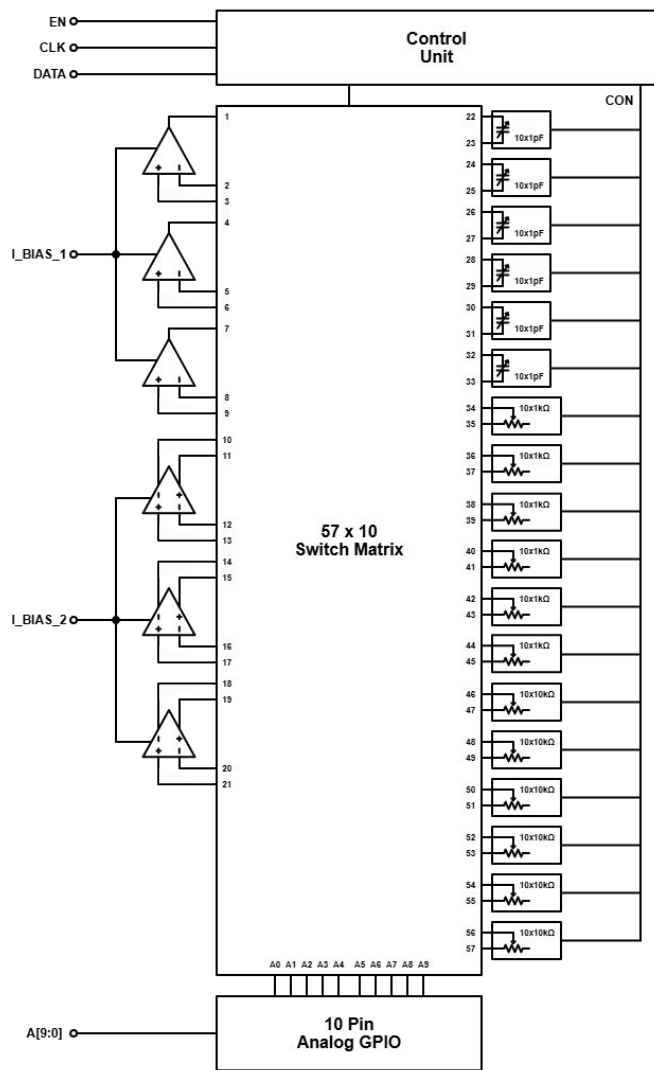
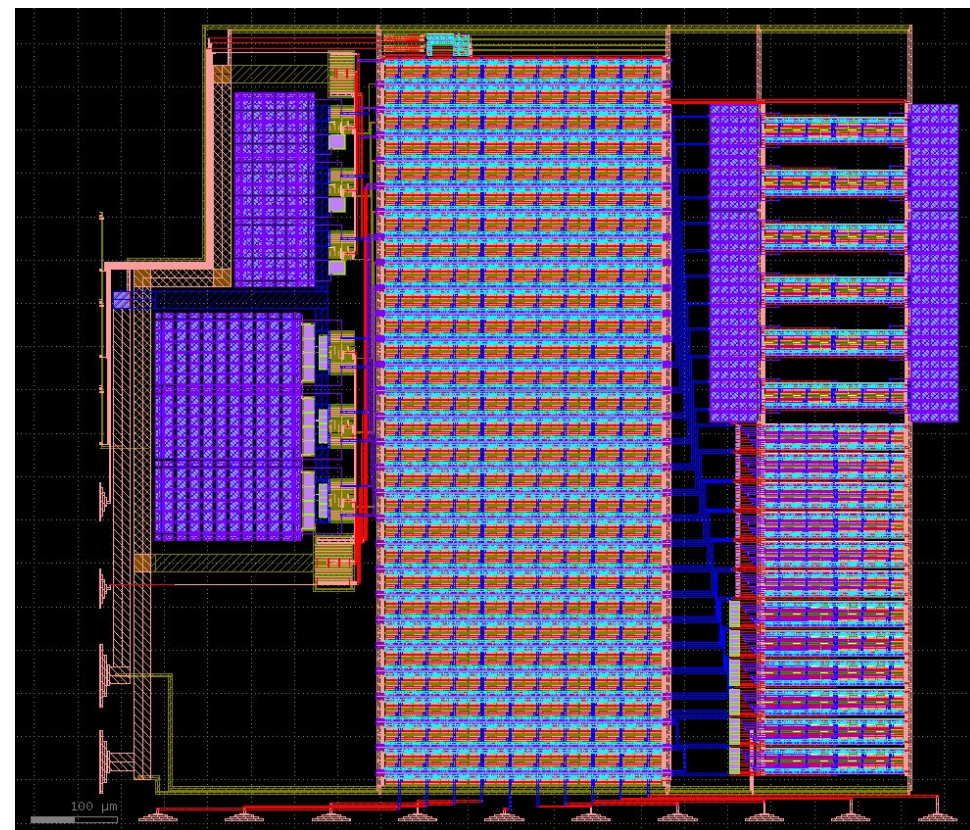


Chipathon 2025 Design Summary

- CreActive (M3) team built a MOSbius-style active filter building blocks consisting of 3 single-ended OTAs, 3 differential output OTAs, 6 capacitor arrays and 12 resistor arrays. Each of the op-amp and passive array terminals are connected to a switch matrix for internal routing and connection to analog GPIO.
- Status:
 - Pre-layout simulation done
 - Integrated inside MOSbius 2 padframe (taped out)
 - DRC clean and block level LVS clean
 - Small attempt on block level PEX and post layout simulation



DATA	73	▶
CLK	72	▶
EN	71	▶
I_BIAS_1	70	▶
I_BIAS_2	69	▶
VDD_M3	68	▶
VSS_M3	67	▶



▲	66	65	64	63	62	61	60	59	58	57	▲
▼	A[9]	A[8]	A[7]	A[6]	A[5]	A[4]	A[3]	A[2]	A[1]	A[0]	▼

Lessons Learned & Best Practices

- What worked well:
 - Using docker as the design environment worked really well without the hassle to set up the tools manually.
 - Adding a team prefix for each circuit block and pin name would help the integration process.
- Improvement needed:
 - LVS tool in Klayout detects capacitor terminal as shorted.
 - Some mismatch in model names between layout and schematic.
 - Klayout DRC check does not provide the complete rule violation.

Reflection & Feedback

- Favorite part
 - The integration step was the most challenging and fun part of this event. We went back and forth revising and verifying as feedback came in. The bigger picture was the whole event brought people with similar interest together and the interaction within the community was a fun learning experience.
- Design tools experience
 - We used Klayout to do the full layout for the first time and it worked surprisingly well for the gf180mcu PDK as we had full control of the layers, as opposed to partially generated layer in magic. The ease of use of magic definitely depends on the maturity of the .mag library.
- Future Suggestions
 - A tailored video on demand tutorial might be a useful addition as the same information can be delivered in shorter time and it can be used for other future events.

Team Profile

- Team name: CreActive
- Team members:
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